

MODULE 02 · DOMAIN 1 - AI & ML FUNDAMENTALS

AWS Managed AI Services

Twelve services, one skill: turning a business sentence into a service name.

DOMAIN 1 of 5 EXAM WEIGHT 20% MODULE 2 of 14

Every wrong option will be a **real AWS service that does something real**. It just will not do **this**.

Know which shelf you are reaching for

01

AI services

Pre-trained. Call an API. No ML skills, no training data, no model to manage.

02

SageMaker AI

Build, train and deploy your own models. Full control, full responsibility.

03

Bedrock

Foundation models via API. Customise with prompts, RAG or fine-tuning.

ON THE EXAM

The words 'no ML expertise' or 'least development effort' point at tier one. 'Custom model' or 'proprietary algorithm' points at SageMaker.

The four you must not confuse

SERVICE	THE VERB IT PERFORMS
Comprehend	Understands the meaning of text. Sentiment, entities, language, PII.
Textract	Extracts text and structure out of scanned documents and forms.
Translate	Converts text from one language into another.
Kendra	Searches your documents and returns a direct answer, not a link list.

ON THE EXAM

Textract reads what the page says. Comprehend works out what it means. That single sentence resolves most confusion between them.

Natural language understanding as an API

- ◆ **Sentiment analysis** — positive, negative, neutral or mixed. The classic use case.
- ◆ **Entity recognition** — pulls out people, places, dates, quantities, organisations.
- ◆ **Key phrase extraction** and **dominant language detection**.
- ◆ **PII detection and redaction** — finds and masks personal data inside free text.
- ◆ **Custom classification** — train it on your own categories with labelled examples.

ON THE EXAM

Customer reviews, support tickets or social posts plus the word sentiment means Comprehend. No exceptions.

Getting data out of documents

- ◆ Reads printed and handwritten text from scans, PDFs and photographs of documents.
- ◆ Understands **structure**: form fields with their values, and table cells with their rows.
- ◆ Purpose-built handling for invoices, receipts and identity documents.
- ◆ It does **not** interpret meaning. It hands you the text and the layout, nothing more.

ON THE EXAM

Plain OCR reads characters. Textract also gives you the form field, its value, and the table structure — that is why it beats generic OCR options.

Textract or Comprehend?

TEXTTRACT

Reads the page

- Input is an image or a scanned PDF
- Output is text, form fields and tables
- Answers: what does this document contain?
- Processing invoices, forms, ID documents

COMPREHEND

Reads the meaning

- Input is text you already have
- Output is sentiment, entities, categories
- Answers: what is this text about?
- Analysing reviews, tickets, survey responses

THE TIEBREAKER

Follow the input. Scanned or photographed document means Textract. Text already in digital form means Comprehend.

Three services, three directions

SERVICE	DIRECTION
Transcribe	Speech in, text out. Call recordings, meetings, media captions.
Polly	Text in, speech out. Announcements, accessibility, voice responses.
Translate	Text in one language, text in another.

ON THE EXAM

Transcribe supports speaker diarisation — labelling who spoke. That detail appears in call-centre questions.

Understanding images and video

- ◆ **Object and scene detection** — labels what appears in an image or video.
- ◆ **Facial analysis and comparison**, plus face search against a collection.
- ◆ **Content moderation** — flags unsafe or inappropriate imagery.
- ◆ **Custom Labels** — train it to recognise your own objects from a small image set.
- ◆ It reads text in images too, but for documents Textract is the correct answer.

ON THE EXAM

Rekognition finds text in a street sign or a product photo. Textract handles structured documents. The word 'form' or 'invoice' always means Textract.

Enterprise search that answers questions

- ◆ Ask a natural language question, get a specific answer extracted from your documents.
- ◆ Connects to S3, SharePoint, Salesforce, Confluence and other repositories.
- ◆ Respects existing access permissions, so users only see what they are allowed to.
- ◆ Semantic, not keyword — it matches intent rather than exact words.

ON THE EXAM

Kendra is intelligent search over documents. It is not a chatbot and it does not generate new text.

Kendra or Comprehend?

KENDRA

Find the answer

- You have a question and a document store
- Returns a passage that answers it
- Employees searching internal policies
- Permission-aware retrieval

COMPREHEND

Analyse the text

- You have text and want properties of it
- Returns sentiment, entities, categories
- Classifying incoming support tickets
- No search involved at all

THE TIEBREAKER

Is someone asking a question? That is Kendra. Are you processing text in bulk to extract properties? That is Comprehend.

Recommendations without building a recommender

- ◆ Trained on **your** user interaction data. Views, clicks, purchases, ratings.
- ◆ Real-time personalisation that adapts as a user's behaviour changes in session.
- ◆ Handles the cold-start problem for new users and new items.
- ◆ Also does related items, personalised ranking, and next-best-action.

ON THE EXAM

Any scenario about suggesting products, content or media to individual users is Personalize. It is rarely a distractor — when it appears, it is usually correct.

Conversational interfaces

- ◆ Builds chatbots and voice bots using **intents**, **utterances** and **slots**.
- ◆ Same technology that powers Alexa. Integrates with Lambda for fulfilment.
- ◆ Structured, task-oriented conversation: booking, ordering, checking a balance.
- ◆ For open-ended generative conversation, the modern answer is Bedrock or Amazon Q.

ON THE EXAM

Lex handles defined tasks with defined steps. If the scenario wants free-flowing generated answers, it is no longer a Lex question.

The ones with narrow, obvious triggers

SERVICE	TRIGGER
Comprehend Medical	Clinical text. Extracts conditions, medications, dosages.
Transcribe Medical	Speech to text for clinical dictation and consultations.
Amazon Q in QuickSight	Ask questions of a dashboard in plain English.
Amazon A2I	Route low-confidence predictions to a human reviewer.
Amazon Forecast	Time-series demand forecasting. See the availability note next.
Amazon Fraud Detector	Online fraud scoring. See the availability note next.

ON THE EXAM

The medical variants only appear when the scenario is explicitly clinical. If it says patient, doctor or diagnosis, take the medical option seriously.

Some of these are closed to new customers

- ◆ **Amazon Forecast** — closed to new customers since July 2024. AWS points to SageMaker Canvas.
- ◆ **Amazon Fraud Detector** — closed to new customers since November 2025.
- ◆ Several **Comprehend** and **Rekognition** features moved to maintenance in April 2026.
- ◆ Learn them anyway. Exam question banks lag behind service changes.

ON THE EXAM

If Forecast or Fraud Detector is the best fit for a described scenario, still choose it. You are being tested on the catalogue as documented, not on procurement reality.

Intelligent document processing

01

Textract

Extract text, form fields and tables from the scanned document.

02

Comprehend

Classify the document and pull out entities or redact PII.

03

A2I

Send anything below the confidence threshold to a human.

04

Store

Write the structured result to S3 or a database.

ON THE EXAM

IDP questions are answered by recognising the chain, not one service. Textract then Comprehend, in that order, with A2I for the uncertain cases.

Human review, built into the pipeline

- ◆ Triggers a human review when a prediction's **confidence score** falls below a threshold.
- ◆ Reviewers can be your own staff, a vendor workforce, or Mechanical Turk.
- ◆ The human decisions become training data for improving the model.
- ◆ Appears in both Domain 1 and Domain 4, where it is the human-oversight answer.

ON THE EXAM

Any scenario asking how to catch uncertain predictions before they reach a customer is Amazon A2I.

Managed AI service or build it yourself?

USE A MANAGED AI SERVICE

The problem is standard

- Sentiment, OCR, translation, transcription
- No ML expertise on the team
- Fastest time to value, lowest effort
- Pay per API call, nothing to operate

BUILD ON SAGEMAKER

The problem is yours alone

- A domain-specific model no service offers
- You have data scientists and labelled data
- You need control of the algorithm
- You accept training and hosting costs

THE TIEBREAKER

'Least operational overhead' and 'no ML expertise' are the managed-service signals.
'Custom algorithm' and 'full control' point at SageMaker.

Scenario to service, at a glance

THE SCENARIO MENTIONS	THE SERVICE IS
Customer reviews, sentiment, tickets	Comprehend
Scanned forms, invoices, handwriting	Textract
Photos, video, faces, moderation	Rekognition
Call recordings, meeting audio, captions	Transcribe
Reading text aloud, accessibility	Polly
Multiple languages, localisation	Translate
Searching internal documents for answers	Kendra
Product or content suggestions per user	Personalize
Booking or ordering chatbot	Lex
Human review of uncertain predictions	A2I

How the wrong answers are built here

- ◆ **Right data, wrong verb.** Comprehend offered for a scanned form, because both involve documents.
- ◆ **Right verb, wrong data.** Rekognition offered for invoice fields, because both extract text.
- ◆ **The heavier option.** Building on SageMaker when a managed service already does it.
- ◆ **The adjacent service.** Kendra offered when the task is classification, not search.

ON THE EXAM

When two options both look plausible, name the exact verb each service performs. The one whose verb matches the scenario wins.

Carry these into the exam

- ◆ Textract reads what the page says. Comprehend works out what it means.
- ◆ Scanned or photographed input means Textract. Already-digital text means Comprehend.
- ◆ Someone asking a question means Kendra. Bulk text processing means Comprehend.
- ◆ Documents and forms means Textract, even though Rekognition can read text.
- ◆ IDP order is Textract, then Comprehend, with A2I for low confidence.
- ◆ 'No ML expertise' and 'least effort' always point away from SageMaker.

A property insurer receives around forty thousand handwritten claim forms by post each month. Staff currently retype the policy number, claim amount and incident date into the claims system.

The company has no machine learning engineers and wants the fastest route to automation.

Which service should the company use?

- | | |
|---|--|
| ☐ A Amazon Comprehend | ☐ B Amazon Textract |
| ☐ C Amazon Rekognition | ☐ D A custom model on SageMaker |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

 Amazon Comprehend
Cannot read a scanned image. Needs digital text as input.
- B

 Amazon Textract
Reads handwriting and returns form fields with their values.
- C

 Amazon Rekognition
Reads text in images, but not structured form fields.
- D

 Custom SageMaker model
The heavier option. No ML engineers, and a service already does this.

ON THE EXAM

Handwritten plus named form fields plus no ML expertise. Three separate signals all pointing at Textract.

A telecoms company records all customer support calls. It wants to know which calls involved an angry customer, and to see the trend by month.

The calls are audio files stored in Amazon S3.

Which combination of services meets the requirement?

- | | |
|--|--|
| ☐ A Transcribe, then Comprehend | ☐ B Comprehend, then Transcribe |
| ☐ C Transcribe, then Kendra | ☐ D Polly, then Comprehend |

◆ PAUSE AND DECIDE

Which combination of services meets the requirement?

- A** Transcribe, then Comprehend
Audio to text, then text to sentiment. Correct order.
- B** Comprehend, then Transcribe
Comprehend cannot accept audio. Order is reversed.
- C** Transcribe, then Kendra
Kendra searches. It does not measure sentiment.
- D** Polly, then Comprehend
Polly generates speech. The company already has speech.

ON THE EXAM

Chain questions are usually decided by order. Only one sequence has each service receiving input it can actually accept.

A large engineering firm has twenty years of technical standards, safety procedures and project reports spread across SharePoint and S3.

Engineers waste hours hunting for specific answers. Access to some documents is restricted by department.

Which service best meets the requirement?

☐ A Amazon Comprehend

☐ B Amazon Kendra

☐ C Amazon Lex

☐ D Amazon Personalize

◆ PAUSE AND DECIDE

Which service best meets the requirement?

- A

 Amazon Comprehend
Analyses text properties. Nobody is asking it a question.
- C

 Amazon Lex
Task-oriented bots with defined intents, not document search.
- B

 Amazon Kendra
Connects to SharePoint and S3, and respects document permissions.
- D

 Amazon Personalize
Recommends items to users. No search capability.

ON THE EXAM

The permissions sentence is doing the work. Kendra is the service that inherits existing document access controls.

A bank automatically extracts data from mortgage applications. Roughly six percent of extractions come back with low confidence scores.

Regulation requires that no application is rejected on the basis of an uncertain automated reading.

What should the bank add to the pipeline?

- ☐ A Amazon A2I
- ☐ B SageMaker Model Monitor
- ☐ C Retrain the model on more data
- ☐ D Amazon Comprehend

◆ PAUSE AND DECIDE

What should the bank add to the pipeline?

- A

 Amazon A2I
Routes low-confidence predictions to a human reviewer.
- B

 SageMaker Model Monitor
Watches for drift over time. Does not review individual cases.
- C

 Retrain on more data
May reduce uncertainty, but never eliminates it. Regulation needs a guarantee.
- D

 Amazon Comprehend
Analyses meaning. Does nothing about confidence thresholds.

ON THE EXAM

Low confidence plus a requirement that a human decides equals A2I. Retraining is the trap because it sounds like real improvement.